

Beyond Assembly: Why India Must Build the Component Layer

India has made significant progress in electronics assembly and system manufacturing. The next challenge is to strengthen the component ecosystem beneath it—from dependable supply and qualification to domestic manufacturing of critical discrete semiconductor devices.

India Is Building Electronics. The Next Challenge Is Building the Components Behind Them

India's electronics industry is entering a new phase. The conversation is no longer limited to assembly. It is increasingly about how much of the value chain India can build, support and sustain within the country.

This is particularly important for discrete semiconductor devices. Diodes, transistors, rectifiers, Schottky and Zener devices, switching components, bridge rectifiers and protection devices are fundamental building blocks across electronic systems.

SR Integrated Circuit India Pvt Ltd is extending its focus from component distribution and sourcing towards developing manufacturing capability for discrete semiconductor devices in India. The objective is to connect product definition, process control, testing, quality, documentation, traceability, inventory and delivery.

Manufacturing Begins Where Sourcing Ends

The biggest difference between sourcing a component and manufacturing one is responsibility for consistency. A sourcing organisation can identify an available product. A manufacturer must demonstrate that the product meets a defined specification and that its characteristics remain controlled from batch to batch.

For discrete semiconductor devices, this includes electrical characteristics, voltage and current ratings, power performance, switching behaviour, thermal characteristics and package dimensions. Quality therefore becomes a process discipline rather than simply a final inspection activity. Testing is most effective when supported by controlled materials, defined process conditions and traceable production records.

Qualification Is Where Manufacturing Meets Engineering

A manufacturer needs to provide clearly defined specifications, samples and technical information so customers can evaluate the device. The final qualification decision remains with the customer's engineering and quality teams.

The manufacturer's role is to provide a product that is clearly defined, consistently manufactured, appropriately tested and adequately documented.

Supply Chain Is Part of Manufacturing

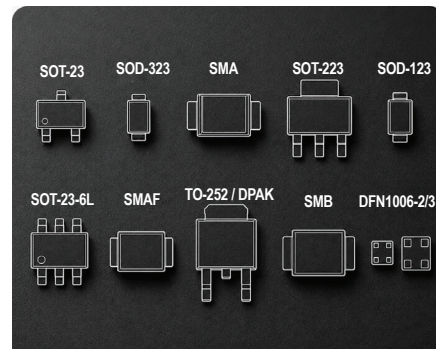
Manufacturing a product and making it reliably available are two different challenges. Once a device enters production, requirements can continue for months or years.

The manufacturing model must connect demand, planning, quality, inventory, traceability, logistics and delivery. Reliable availability is as important as the product itself.

From Distribution to Manufacturing

Distribution provides visibility into manufacturers' requirements and supply gaps. A component may be available today and difficult to source tomorrow. Engineering teams therefore need confidence that electrical and mechanical characteristics remain consistent during production.

SR Integrated Circuit India Pvt Ltd is working towards building manufacturing capability for discrete semiconductor devices. The goal is not simply to produce components, but to create a connected model where product definition, manufacturing, quality, testing, qualification, documentation, traceability, inventory and customer delivery work together.



India's Opportunity

India's electronics opportunity will increasingly depend on what happens beneath the finished product. Assembly, product design, semiconductor fabrication and advanced packaging are important, but they depend on an ecosystem of components that must be available, qualified, documented and supported.

Strengthening this layer does not mean that every component must be manufactured domestically. It means building greater capability, resilience and choice while integrating with global supply chains where appropriate.

Dependable access to qualified components can become a strategic advantage. The next step is to strengthen the component layer that makes modern electronics possible.

About SR Integrated Circuit India Pvt Ltd and ComponentBuy

Through its ComponentBuy platform, SR Integrated Circuit India Pvt Ltd is moving towards building capability in discrete semiconductor manufacturing.

ComponentBuy

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Website: www.componentbuy.com